Basic Principles of Medicine II

Module: Nervous System and Behavioral Science | Lecture NB 14

Infratemporal & Pterygopalatine Fossae

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Recommended reading

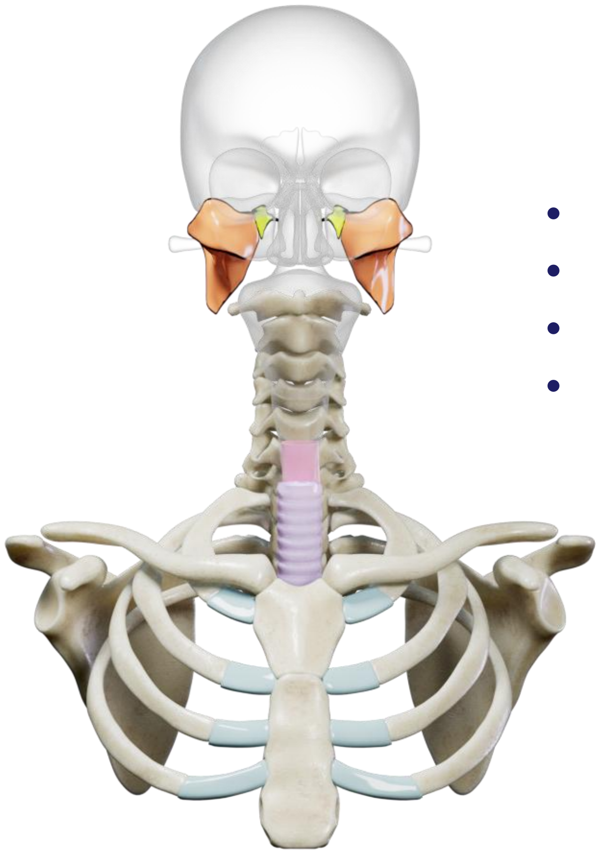
Drake RL, Vogl AW, Mitchell AWM. Gray's anatomy for students. 5th ed. Elsevier. Chapter 8.

Pay special attention to the 'In the Clinic' boxes which contain the clinical correlates for this topic.

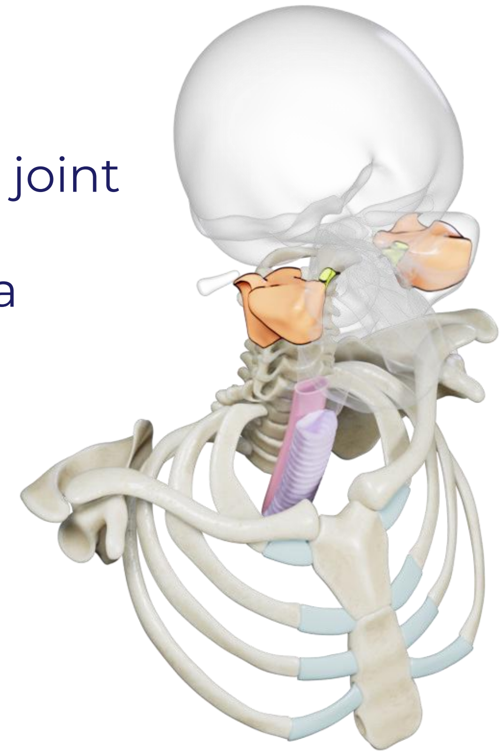
Objectives

SOM.MKII.BPM2.1.NB.1.ANAT.0142	List the bony boundaries of the infratemporal fossa
SOM.MKII.BPM2.1.NB.1.ANAT.0143	Describe the structure, innervation and movements of the temporomandibular joint
SOM.MKII.BPM2.1.NB.1.ANAT.0144	Describe the general attachments, innervations, and actions of the muscles of mastication
SOM.MKII.BPM2.1.NB.1.ANAT.0145	Describe the course of the maxillary artery in the infratemporal fossa and the areas supplied by its branches
SOM.MKII.BPM2.1.NB.1.ANAT.0146	Describe the pterygoid venous plexus and list its intra- and extra-cranial connections
SOM.MKII.BPM2.1.NB.1.ANAT.0147	Describe the mandibular nerve, giving its entrance foramen into the fossa, its branches, the nerve fiber content and distribution of each branch
SOM.MKII.BPM2.1.NB.1.ANAT.0148	Describe the chorda tympani nerve, its origin, fiber types, connection to the lingual nerve, target ganglion and final distrubtion
SOM.MKII.BPM2.1.NB.1.ANAT.0149	Describe the typical site of injection of anesthetic for dental work of the mandibular and maxillary teeth
SOM.MKII.BPM2.1.NB.1.ANAT.0150	Describe the boundaries of the pterygopalatine fossa
SOM.MKII.BPM2.1.NB.1.ANAT.0151	Describe the foramina and connections of the pterygopalatine fossa
SOM.MKII.BPM2.1.NB.1.ANAT.0152	Describe the contents of the pterygopalatine fossa and the relationship of these structures to each other
SOM.MKII.BPM2.1.NB.1.ANAT.0153	Describe the pterygopalatine ganglion, the nerve fibers contributing to its formation, its branches and their distribution
SOM.MKII.BPM2.1.NB.1.ANAT.0154	List the fiber types in the nerve of the pterygoid canal and describe which nerves join to form it
SOM.MKII.BPM2.1.NB.1.ANAT.0155	Describe the pterygopalatine portion of the maxillary artery
SOM.MKII.BPM2.1.NB.1.ANAT.0156	Describe the course and distribution of the sphenopalatine and infraorbital nerves
SOM.MKII.BPM2.1.NB.1.ANAT.0228	Identify major anatomical structures of the infratemporal and pterygopalatine fossa in MRI, CT, and X-ray imaging

Overview



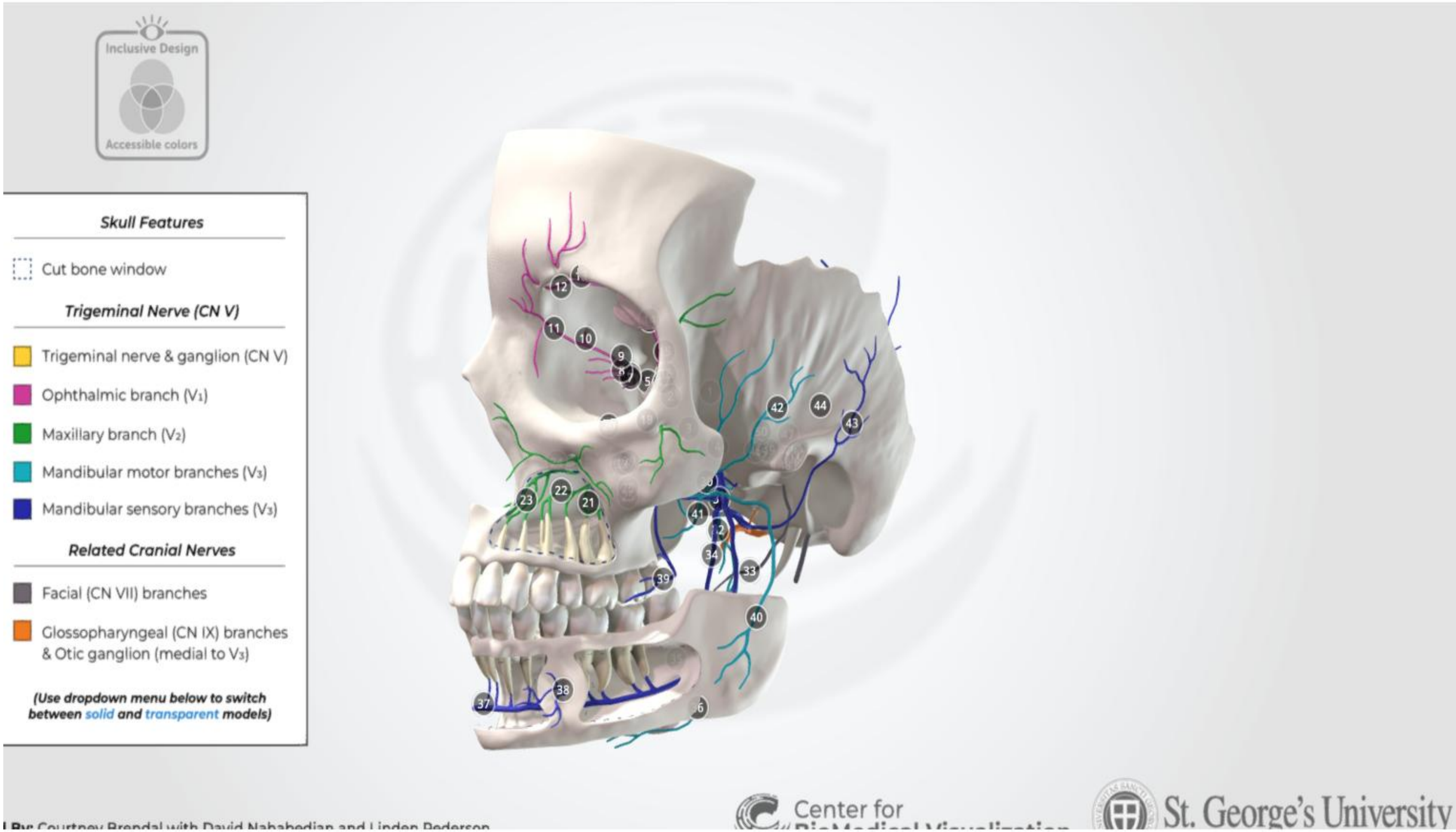
- Temporomandibular joint
- Infratemporal fossa
- Pterygopalatine fossa
- *Cranial nerves:*
  V, VII, IX



Infratemporal Fossa

Pterygopalatine Fossa

3-D Model of IT & PT Fossa Nerves

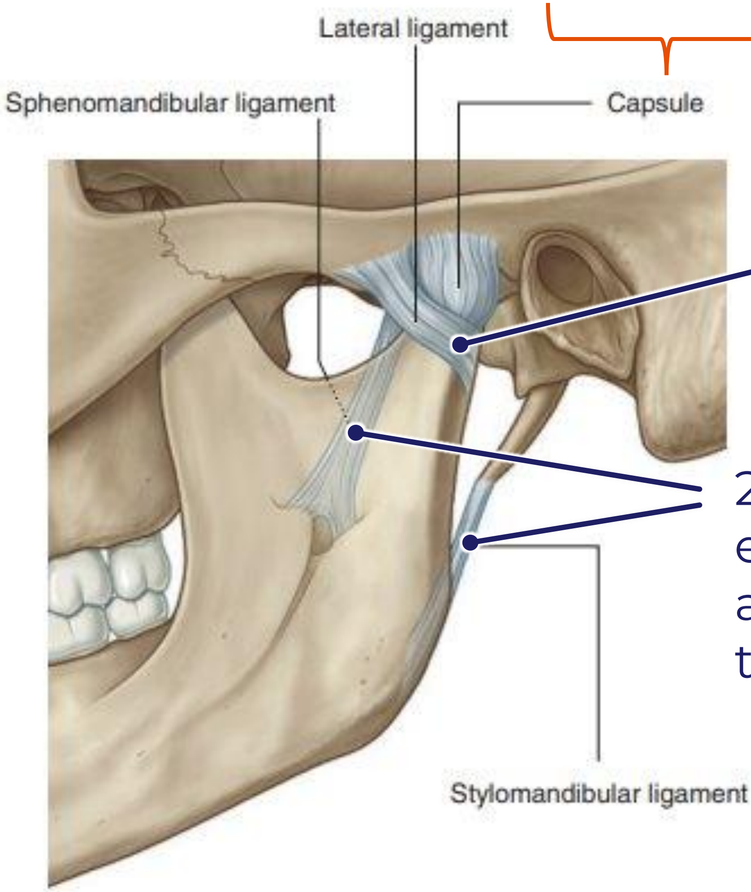


[Nerve Content of the IT & PT Fossae - 3D model by The Center for BioMedical Visualization at SGU (@SGUMedArt) (sketchfab.com)]

Temporomandibular joint (TMJ)

Articulation of the temporal bone with the head of the condyle of the mandible.

The joint is enclosed in a capsule & contains a dense fibrous articular disc

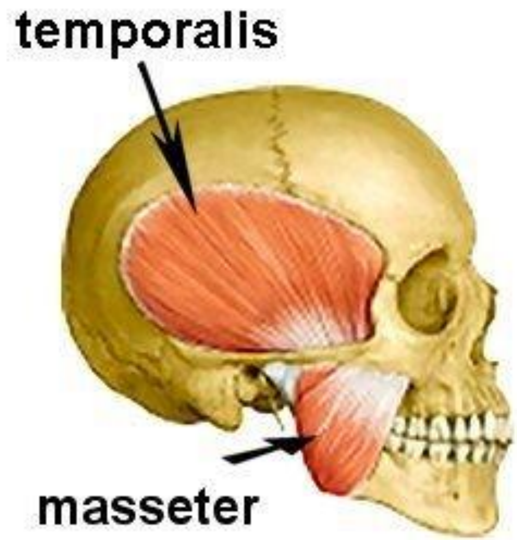


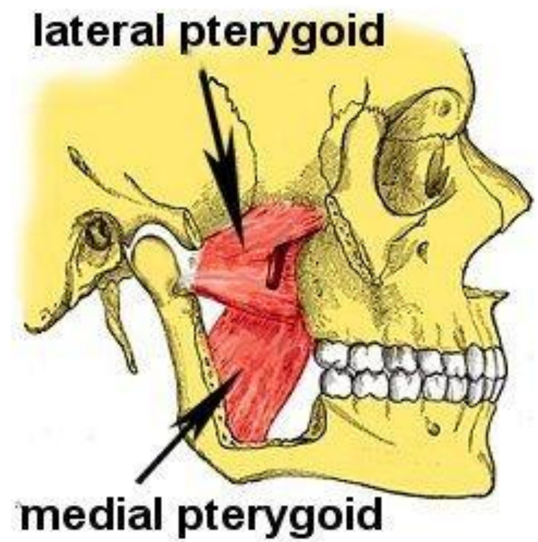
- thickened part of the joint capsule
- strengthens the joint laterally
- prevents posterior dislocation of the joint.

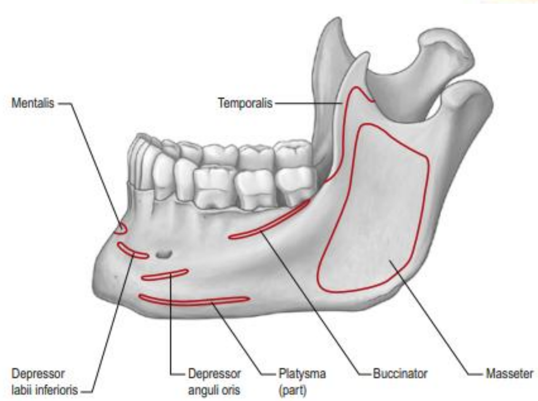
2 supporting extrinsic ligaments attach the mandible to the cranium

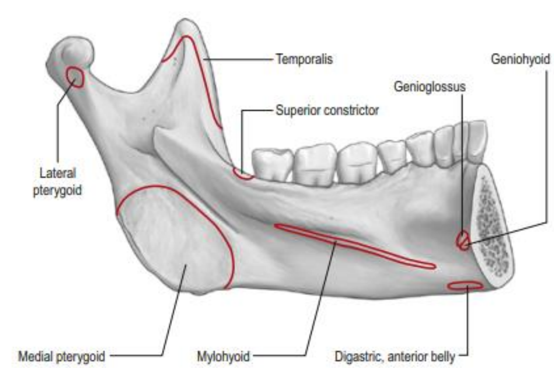
Muscles of Mastication

- TMJ movements are produced chiefly by the muscles of mastication
  - the four muscles develop from the mesoderm of the embryonic first pharyngeal arch.
  - innervated by the nerve of that arch: mandibular nerve (V_3)







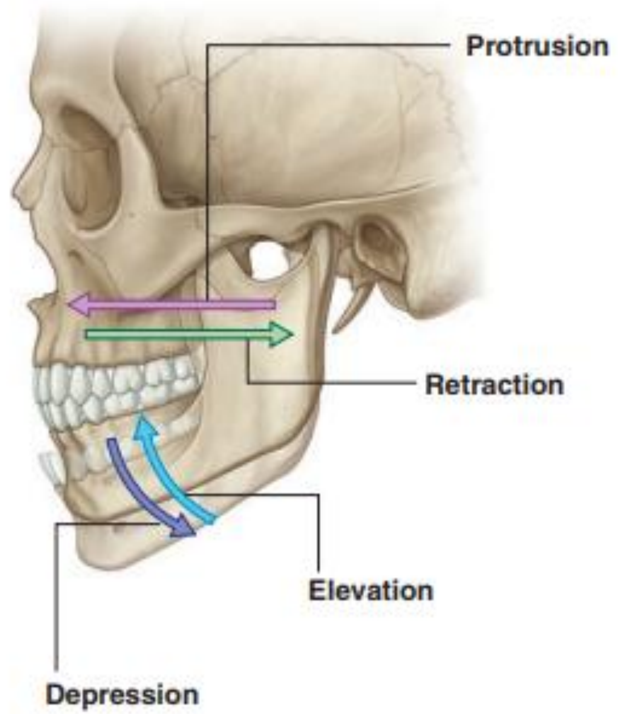


Movements at TMJ

- Muscles of mastication attach to the mandible & produce movements of the lower jaw at the TMJ

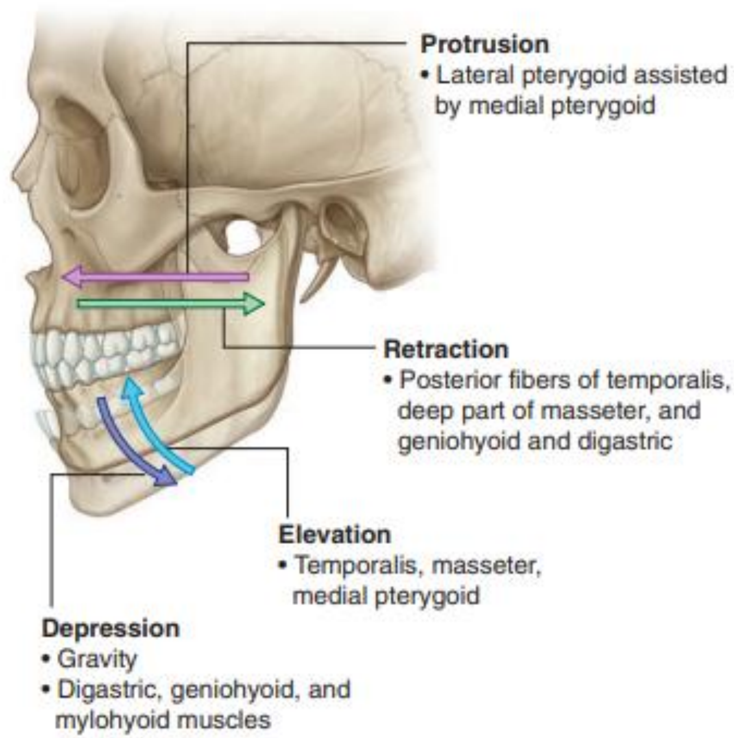
These movements include:

- Retraction & Protrusion (protraction)
- Elevation & Depression
- Rotation: produces side-to-side movements of the mandible



TMJ Functionality

- Muscles of mastication
  - enable chewing and grinding of food
- Arterial supply
  - branches of external carotid (**superficial temporal artery**)
- Sensory innervation from small branches of two nerves
  - *Auriculotemporal nerve*
  - *Masseteric nerve*



TMJ Pain

Encompasses three main entities:

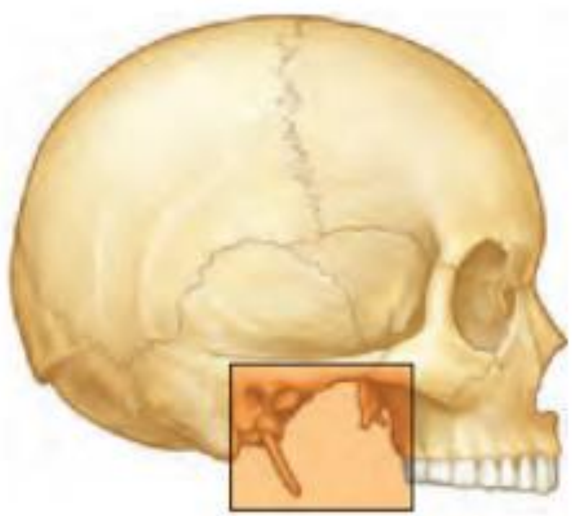
- Dysfunction: the disc is displaced and does not maintain a normal relationship to the articular surfaces throughout the normal range of motion
- Inflammation
  - e.g. rheumatoid arthritis, gout, psoriatic arthritis, other auto-immune disorders
- Trauma: fractures and dislocations

Signs and symptoms of TMJ disorders can include:

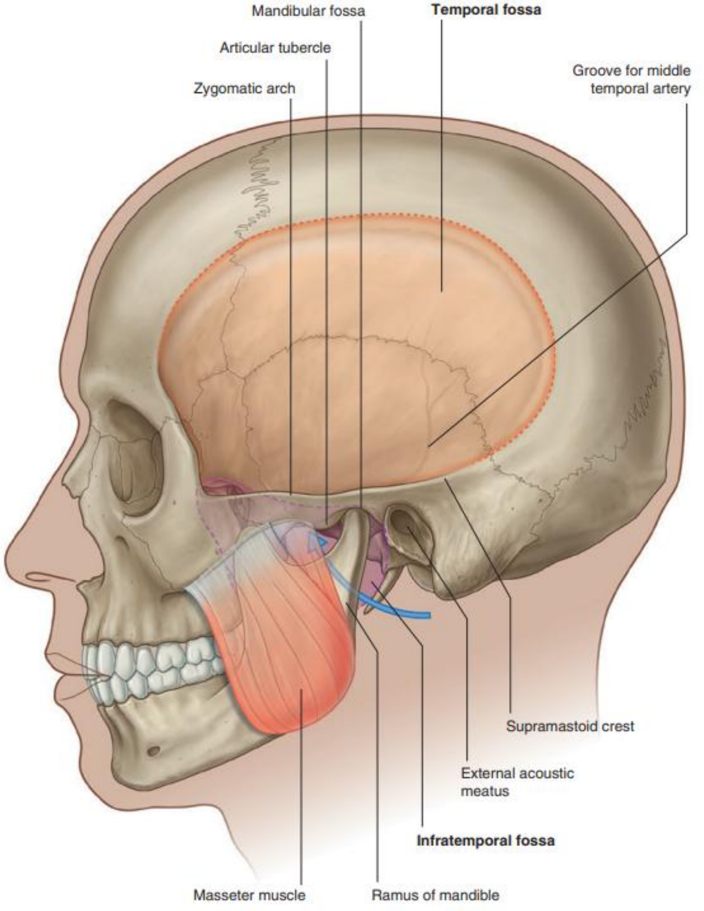
- Pain or tenderness of the jaw
- Pain in one or both TMJs
- Difficulty chewing or pain while chewing
- Aching pain in and around your ear
- Aching facial pain
- Locking of the joint that makes it difficult to open or close the mouth

Infratemporal Fossa

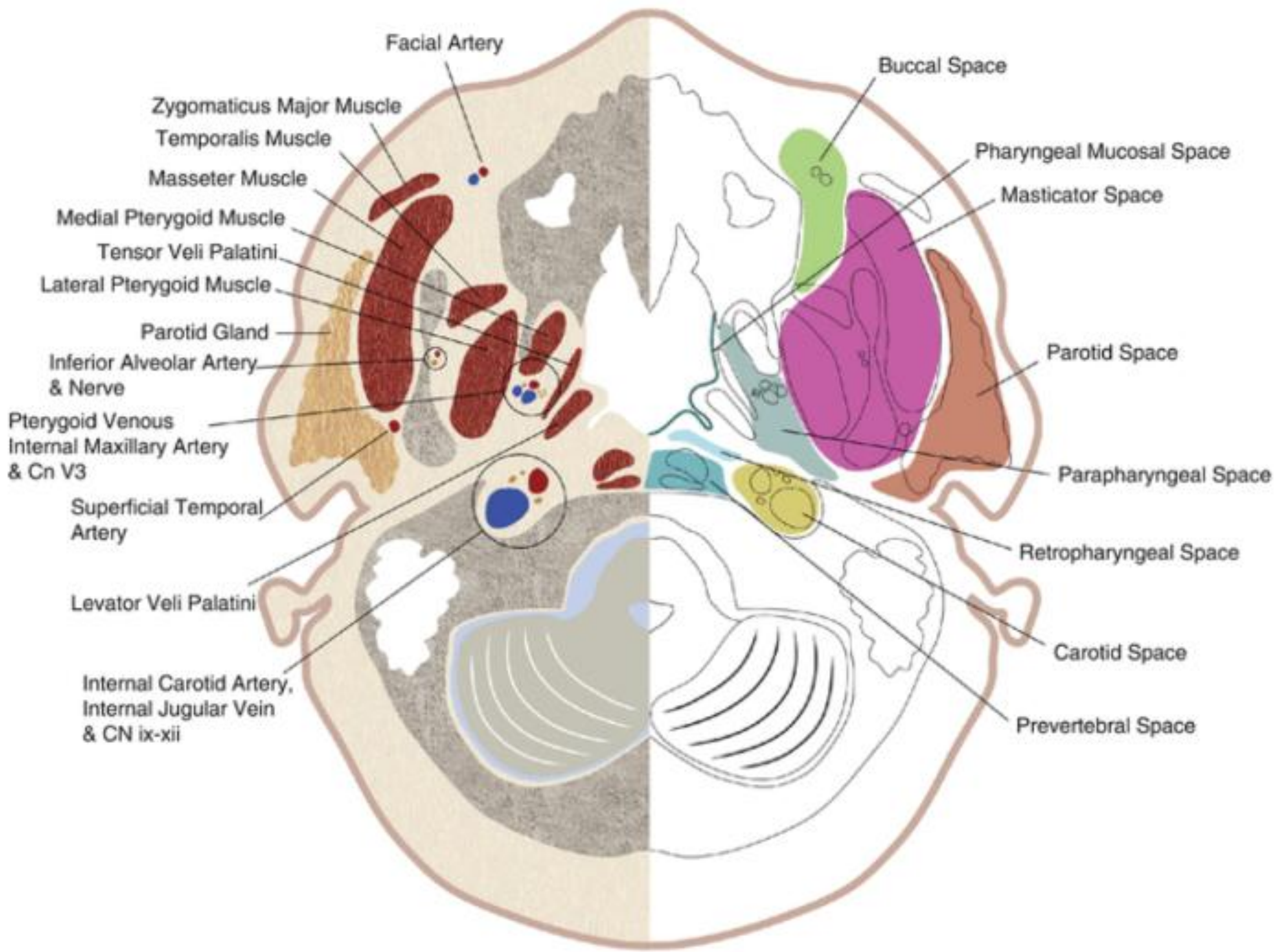
- An irregularly shaped space deep and inferior to the zygomatic arch, deep to the ramus of the mandible, and posterior to the maxilla
- Communicates with the temporal fossa through the interval between (deep to) the zygomatic arch and (superficial to) the cranial bones
- Overlaps with
  - Masticator space
  - Prestyloid (ant) parapharyngeal space



Infratemporal fossa

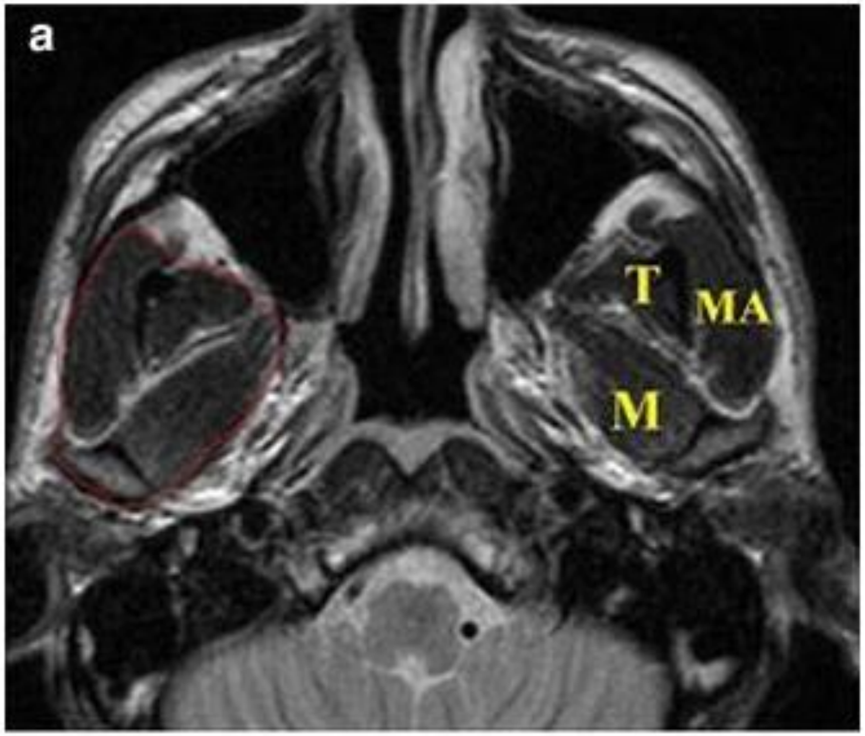


Masticator & Ant Parapharyngeal Space

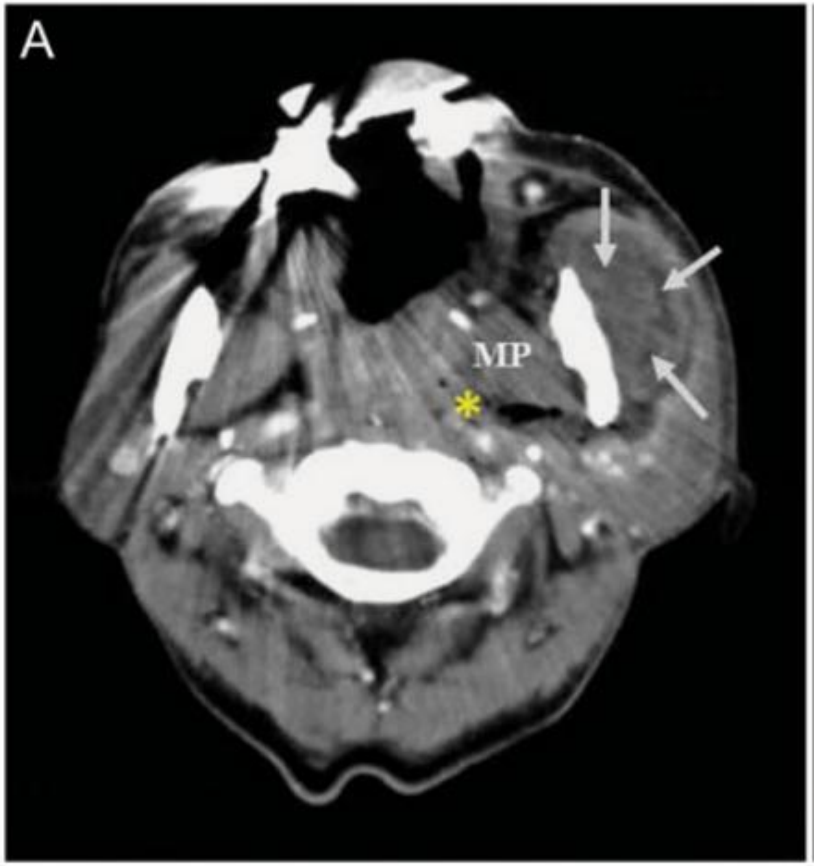


https://radiologykey.com/biopsy-of-head-and-neck-lesions/

Clinical Imaging



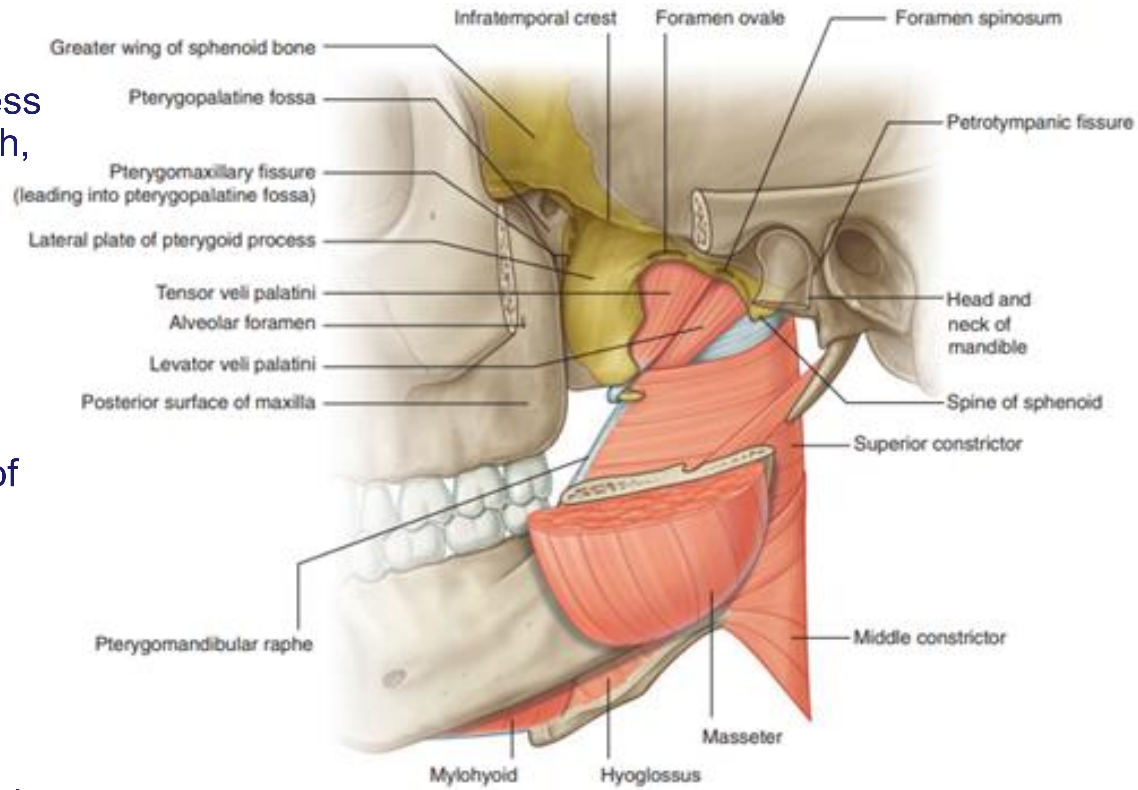
DOI: 10.1007/s11604-014-0289-x



DOI: 10.1177/0003489414526360

Boundaries of IT Fossa

- Medial
  - Lateral pterygoid plate
  - Lateral wall of nasopharynx: tensor veli palatini, levator veli palatini muscles
- Lateral
  - Ramus and coronoid process of mandible, zygomatic arch, temporalis muscle
- Anterior
  - Posterolateral wall of the maxillary sinus
- Posterior
  - Carotid sheath, deep part of the parotid gland, styloid process and its muscles
- Floor
  - Medial pterygoid muscle
- Roof
  - Middle cranial fossa: formed by greater wing of sphenoid (with foramen ovale and foramen spinosum as connections to IT fossa)

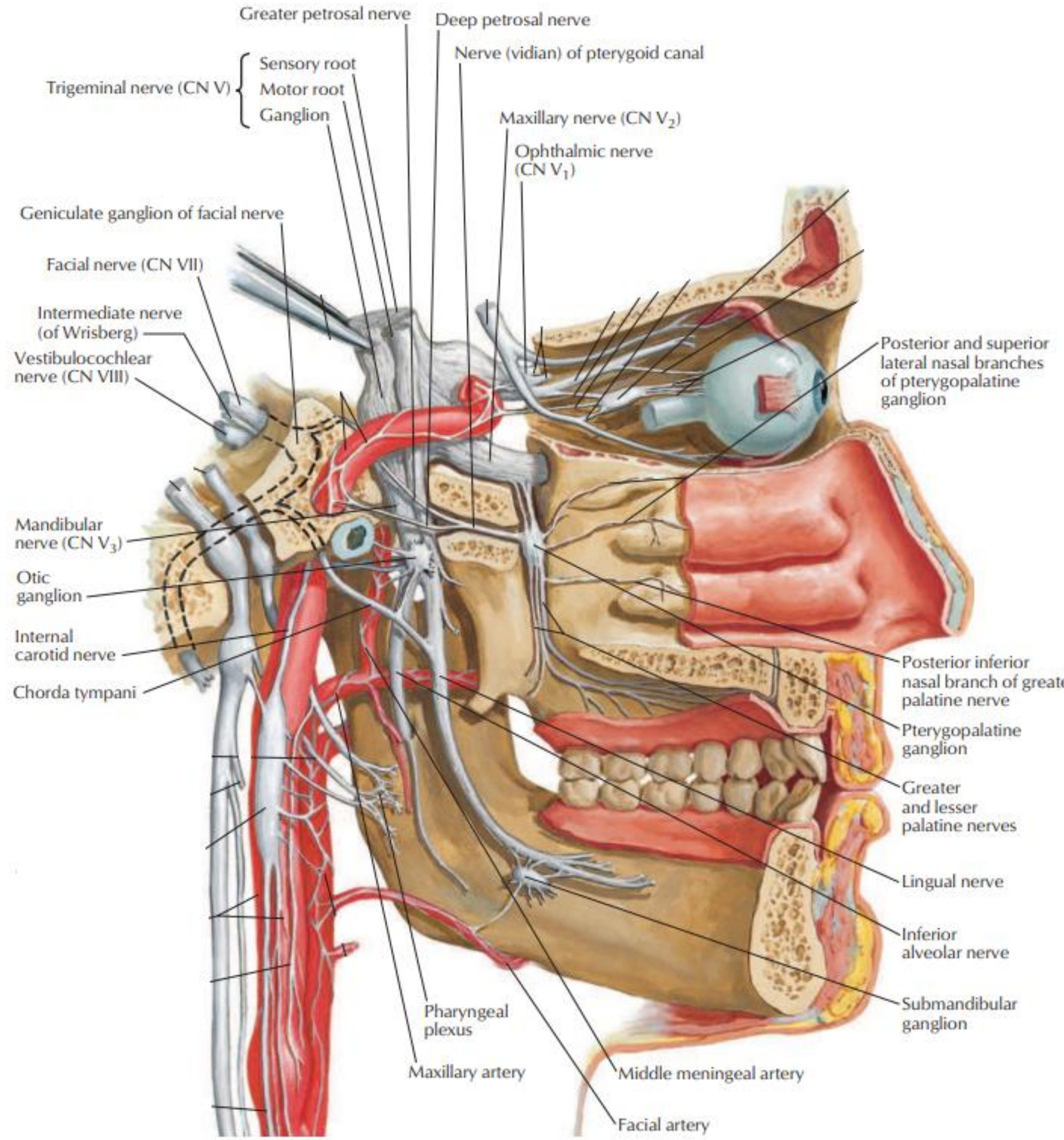


Contents of IT Fossa

- Inferior part of the temporalis muscle
- Lateral and medial pterygoid muscles

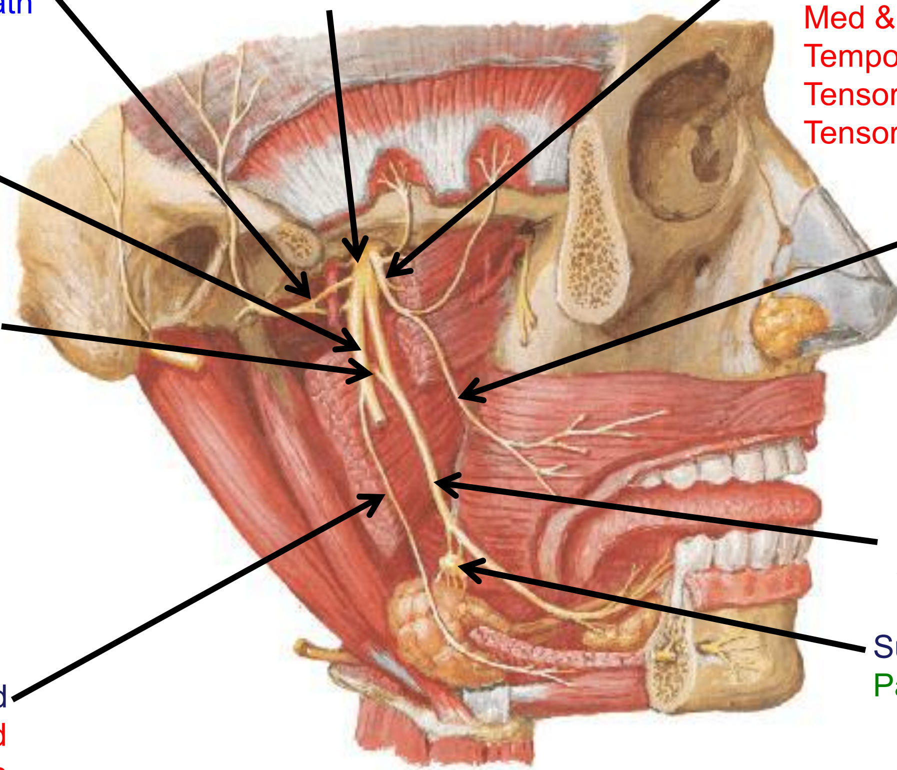
- Pterygoid venous plexus
- Maxillary artery and branches

- Mandibular nerve and its branches
- Chorda tympani nerve
- Otic ganglion

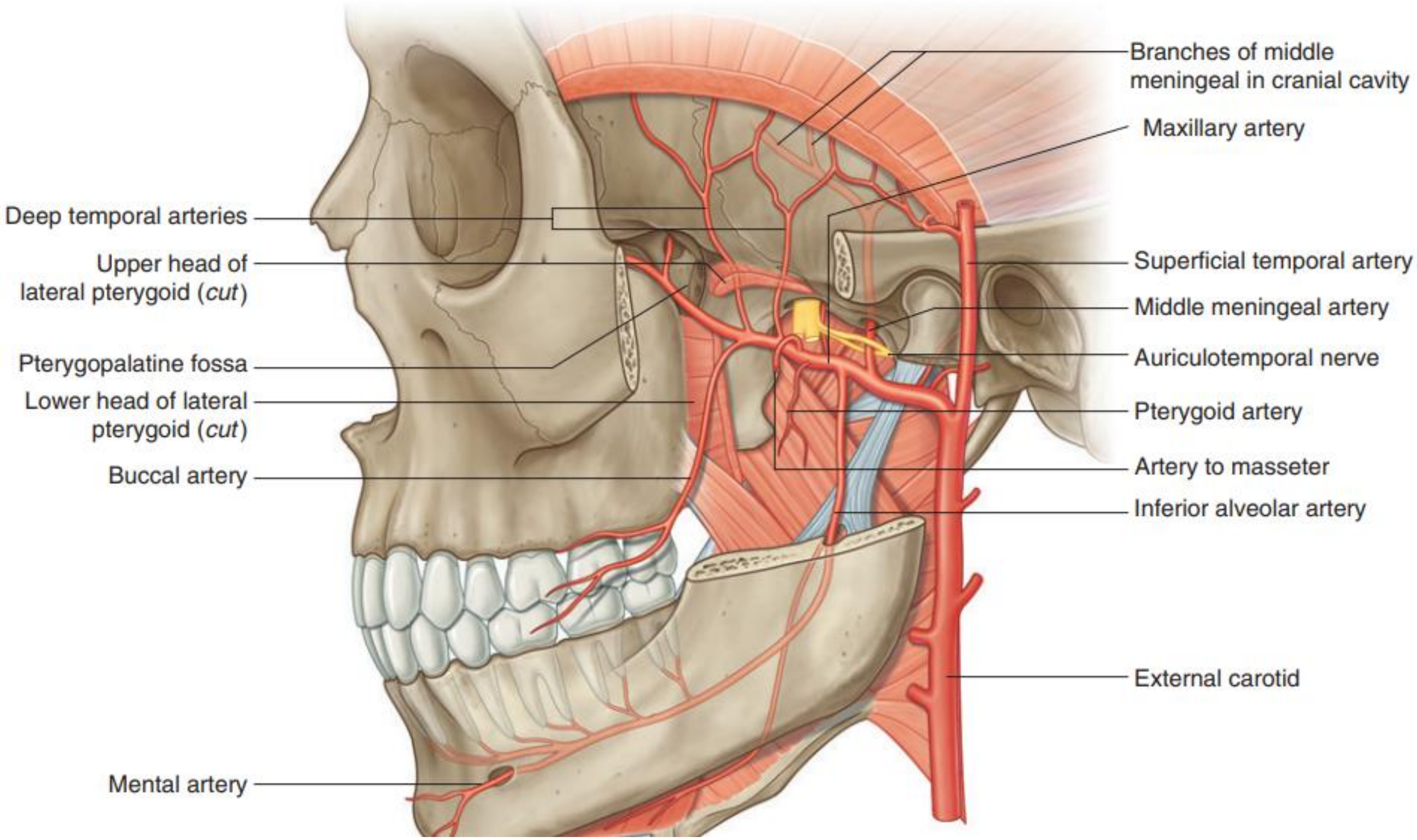


Nerves of the IT fossa

Red: Motor
Blue: Sensory
Green: Parasympathetic



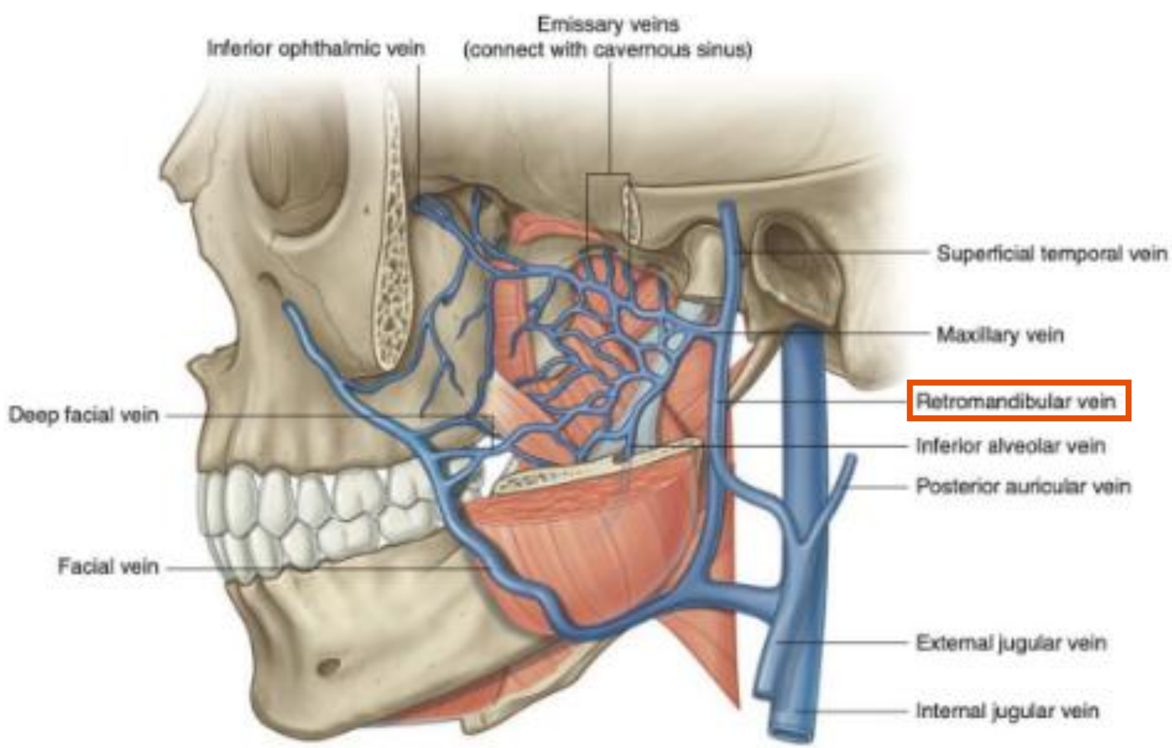
Vasculature of IT Fossa



- **Maxillary artery**: prominent branches include the middle meningeal artery, inferior alveolar artery, deep temporal arteries, and buccal artery
- **Pterygoid venous plexus**: drains the eye and is directly connected to the cavernous sinus
- **Maxillary and Middle meningeal veins**

Pterygoid Plexus

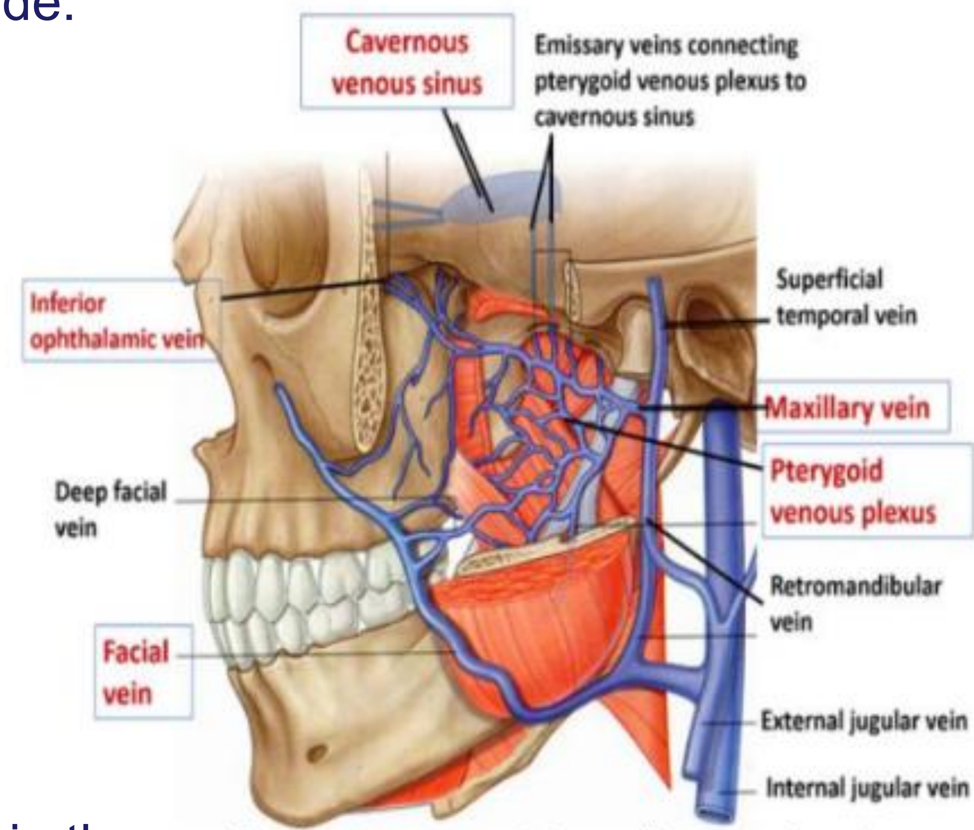
- Connects to the cavernous sinus and eventually drains into the maxillary vein
- Infectious processes originating in the IT fossa have the potential to spread via this venous plexus to the cavernous sinus
  - Can result in a **cavernous sinus thrombosis**



Pterygoid Plexus Tributaries

Tributaries correspond to the branches of the maxillary artery. They include:

- Sphenopalatine
- Deep temporal
- Pterygoid, masseteric
- Buccal, alveolar
- Greater palatine
- Inferior ophthalmic
- Middle meningeal veins



https://anatomyqa.com/pteryygoid-venous-plexus/

Pterygoid plexus anastomoses:

- Anteriorly with the **facial vein** via the deep facial vein
- Superiorly with the **cavernous sinus** via emissary veins

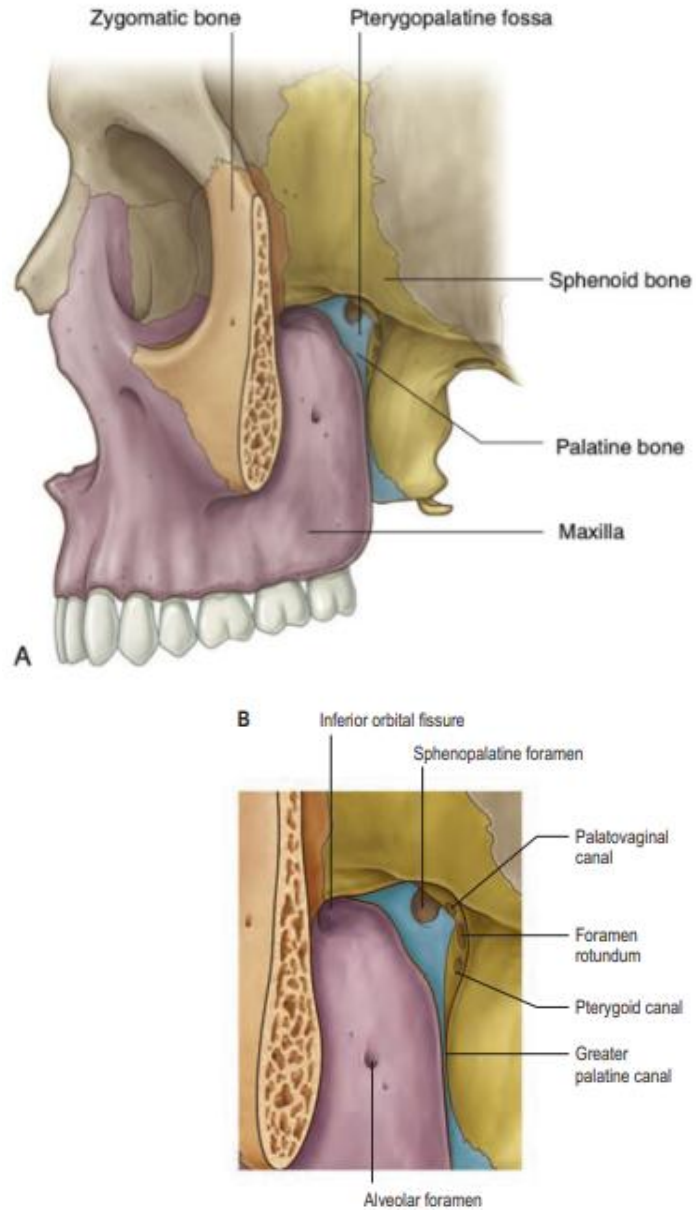
Pterygopalatine (Sphenopalatine) Fossa

Small pyramidal space inferior to the apex of the orbit and medial to the IT fossa

Neurovascular crossroad of:

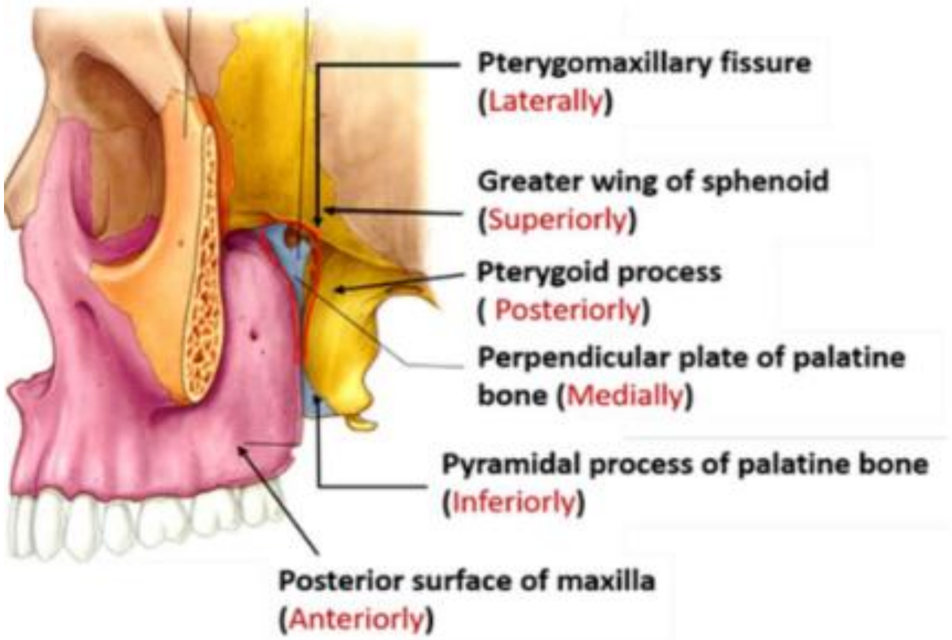
- nasal cavity
- masticator space
- orbit
- oral cavity
- middle cranial fossa

Potential pathway for the spread of neoplastic and infectious processes



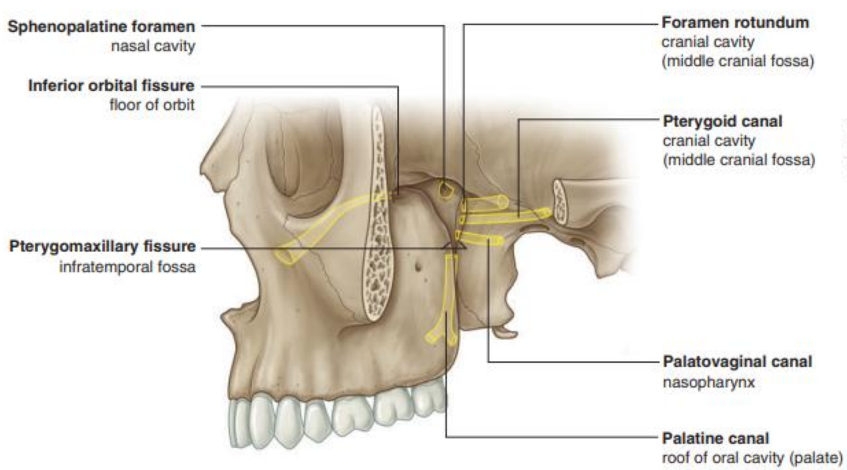
Boundaries of PT Fossa

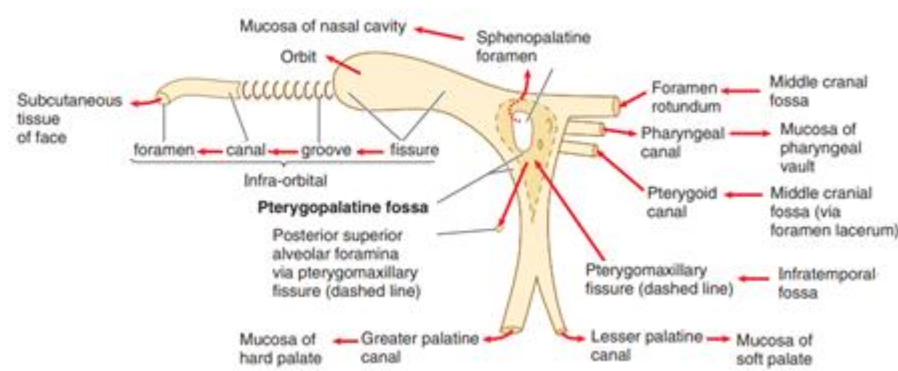
- Medial
  - Perpendicular plate of the palatine bone
- Lateral
  - Narrowing to the pterygomaxillary fissure
- Posterior
  - Pterygoid process of the sphenoid bone
- Anterior
  - Posterior wall of the maxillary sinus
- Roof
  - Body of the sphenoid bone and orbital process of palatine bone
- Floor
  - Pyramid process of the palatine bone



PT Fossa Communication

- Medial: Communicates with the nasal cavity via the **sphenopalatine foramen** (sphenopalatine artery and nasopalatine nerve)
- Lateral: Communicates with the masticator space (or IT fossa) via the **pterygomaxillary fissure**
- Anterosuperior: Communicates with the orbit via the **inferior orbital fissure**
- Posterosuperior: Communicates with Meckel's cave and cavernous sinus via the **foramen rotundum**
- Posteromedial: Communicates with the nasopharynx via the **palatovaginal (pharyngeal) canal**: pharyngeal nerve, pharyngeal branch of maxillary artery
- Posteroinferior: Communicates the **Vidian (pterygoid) canal**: Vidian nerve, artery and vein
- Inferior: Communicates with the palate via **the greater and lesser palatine canals**





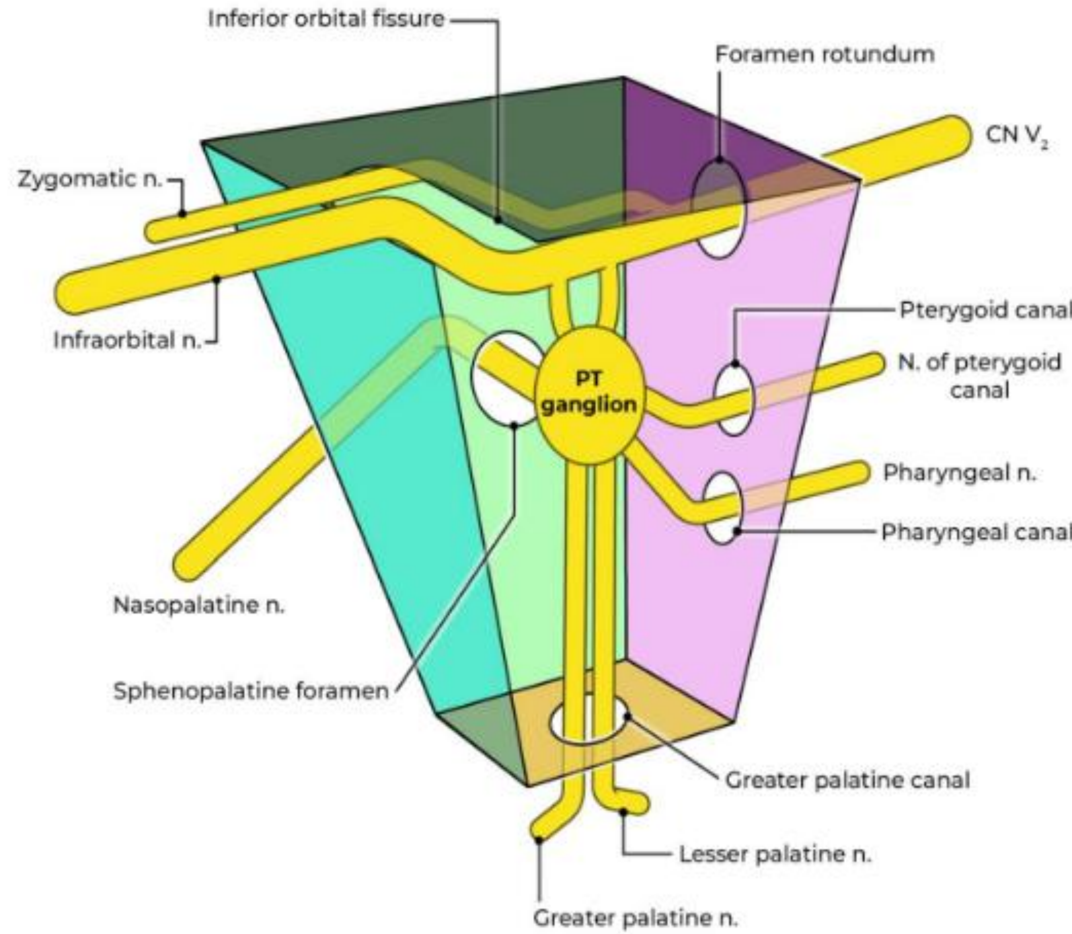
PT Fossa - Model

The pterygopalatine fossa is an inverted, pyramidal-shaped recess on the lateral sides of the skull between the infratemporal (IT) fossa and the nasopharynx.

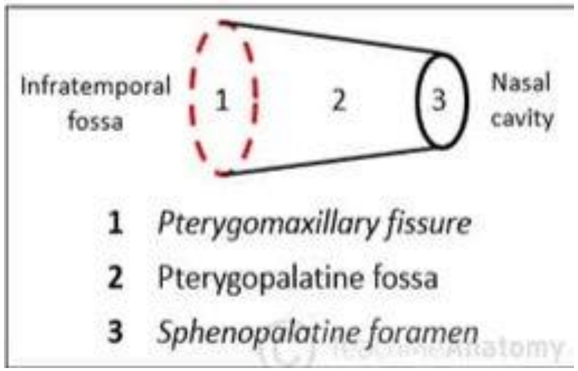
Its location serves as a crossroad for several neurovascular structures.

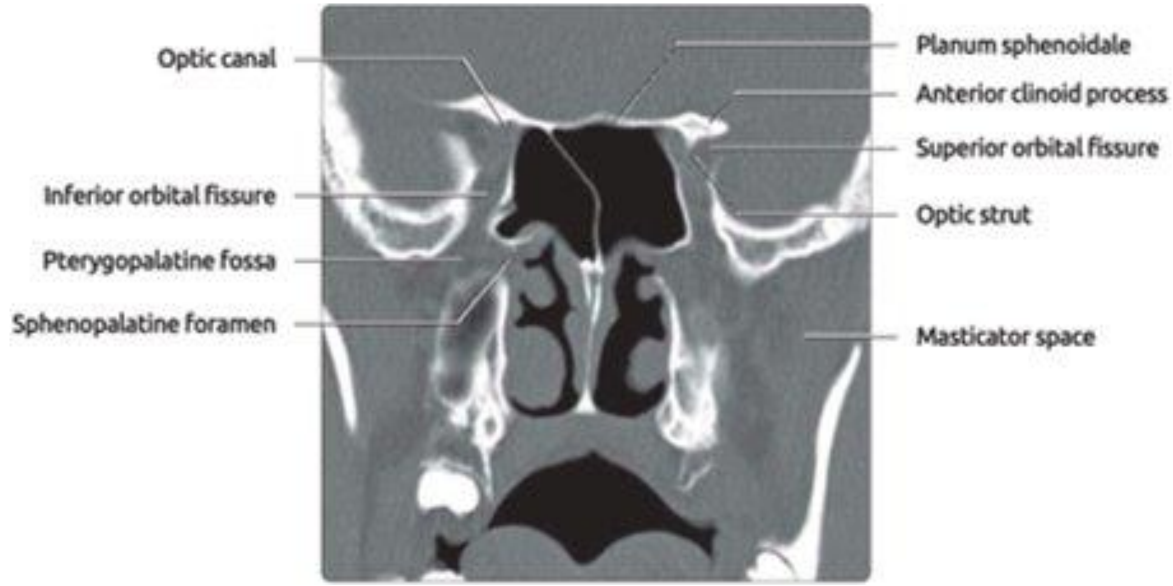
- Anterior: Posterior side of maxilla
- Posterior: Pterygoid process of sphenoid bone
- Superior/Roof: Greater wing of sphenoid bone (infratemporal surface)
- Inferior/Floor: Palatine bone
- Medial: Perpendicular plate of palatine bone

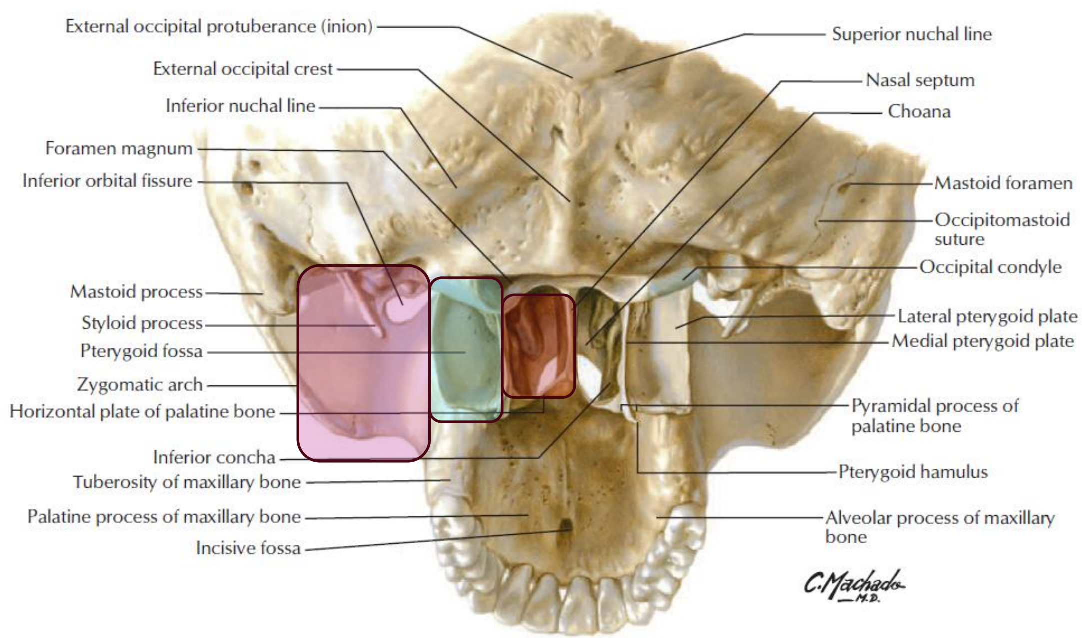
The lateral side is the **pterygomaxillary fissure**, which is an open space that continues into the infratemporal fossa.



IT/PT Fossa - Imaging

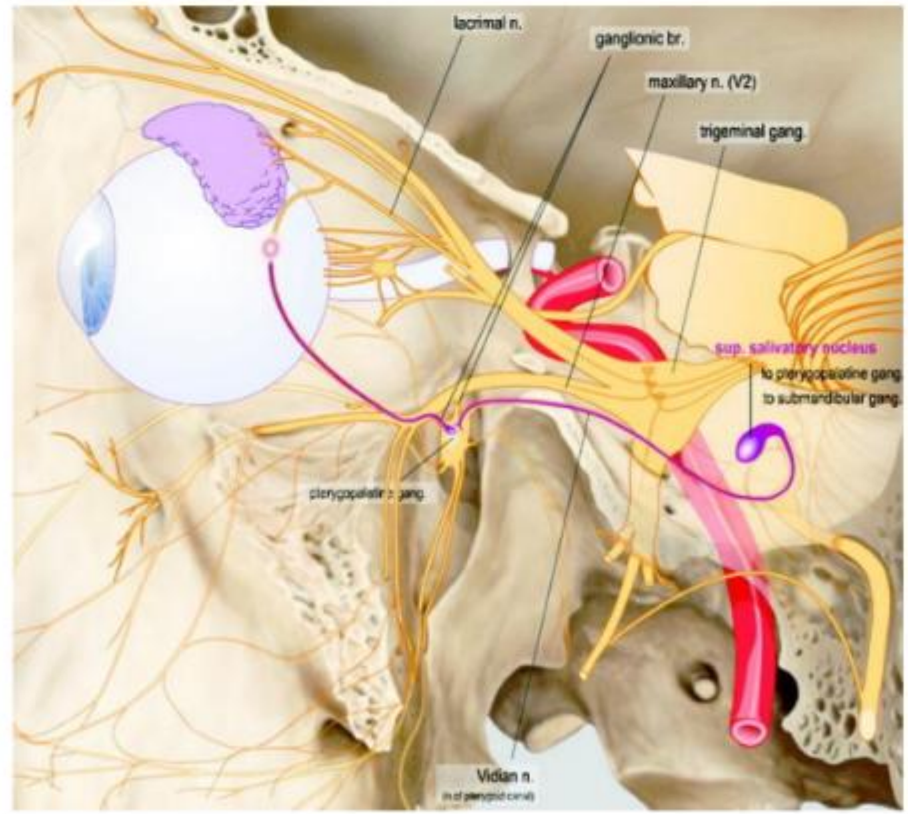






Contents of PT Fossa

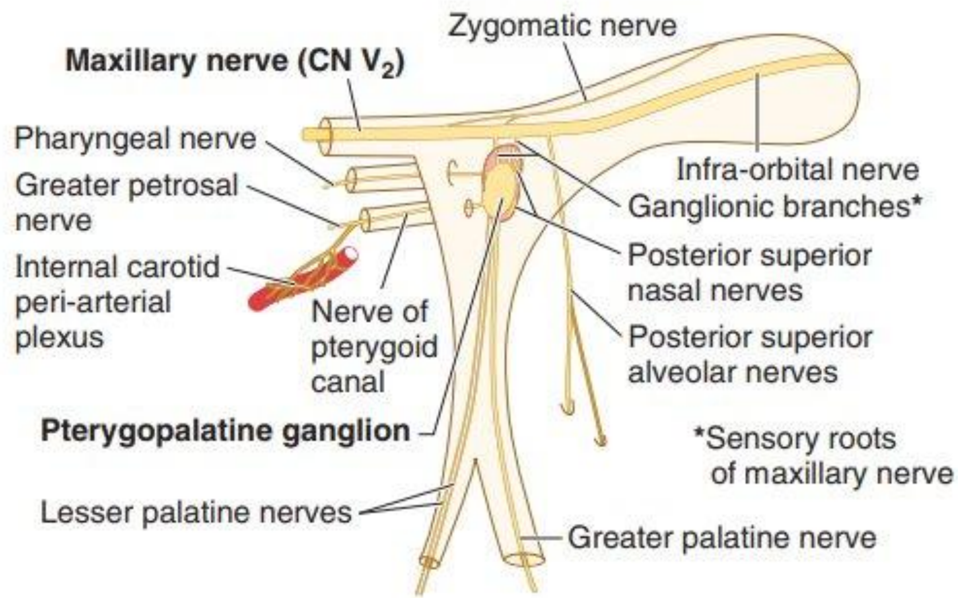
- Maxillary division of trigeminal nerve
  - enters via foramen rotundum
- Pterygopalatine ganglion
- Nerve of the pterygoid canal
- Maxillary artery (terminal portion)
  - branches including the descending palatine artery
- Emissary veins

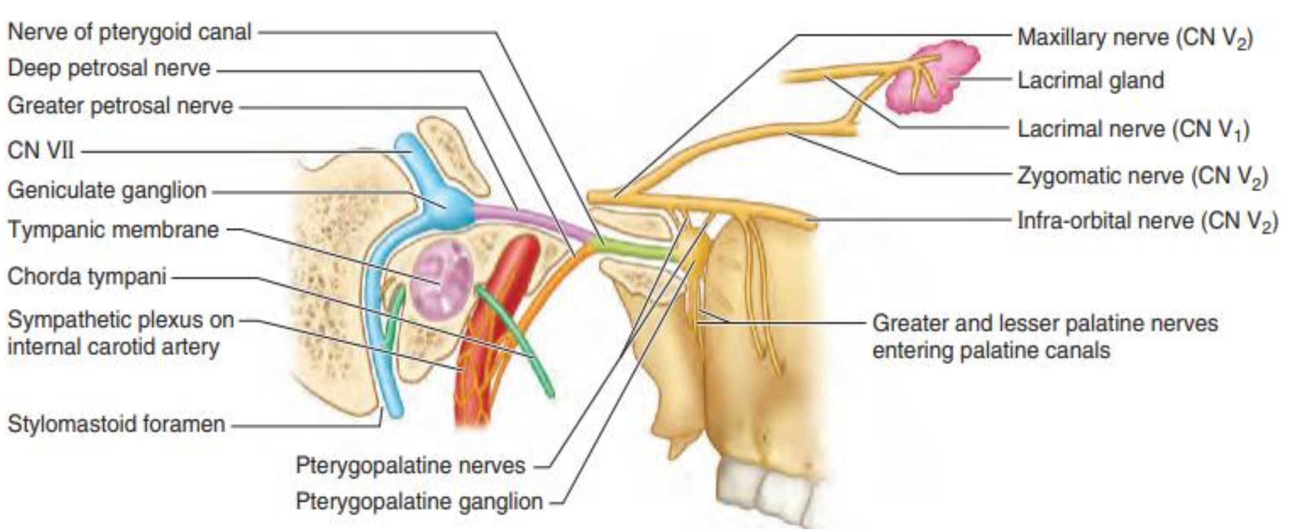


Maxillary nerve – branches

Zygomatic

- Divides into zygomaticotemporal and zygomaticofacial nerves
- Zygomaticotemporal branch carries **parasympathetic secretomotor fibers** to the lacrimal gland via the *lacrimal nerve*





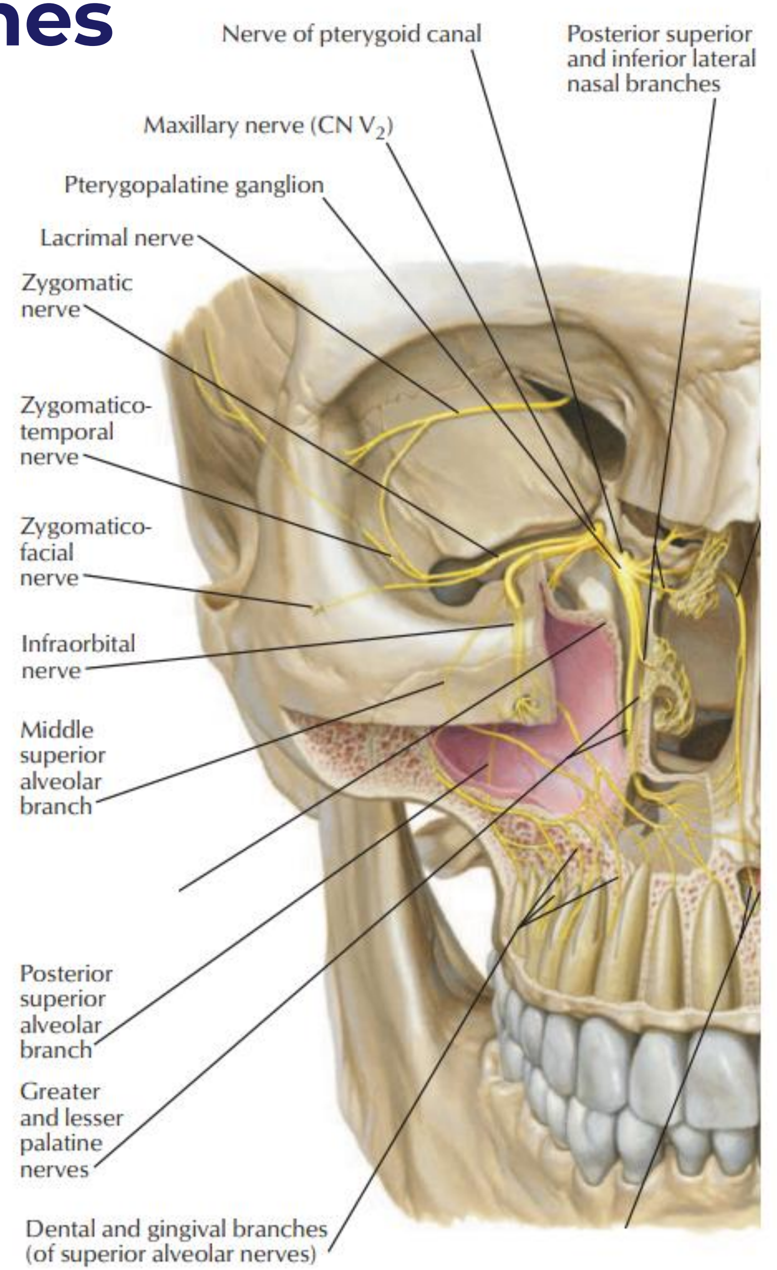
Maxillary nerve – branches

Ganglionic branches

- Nerves that suspend and communicate with the pterygopalatine ganglion

These include

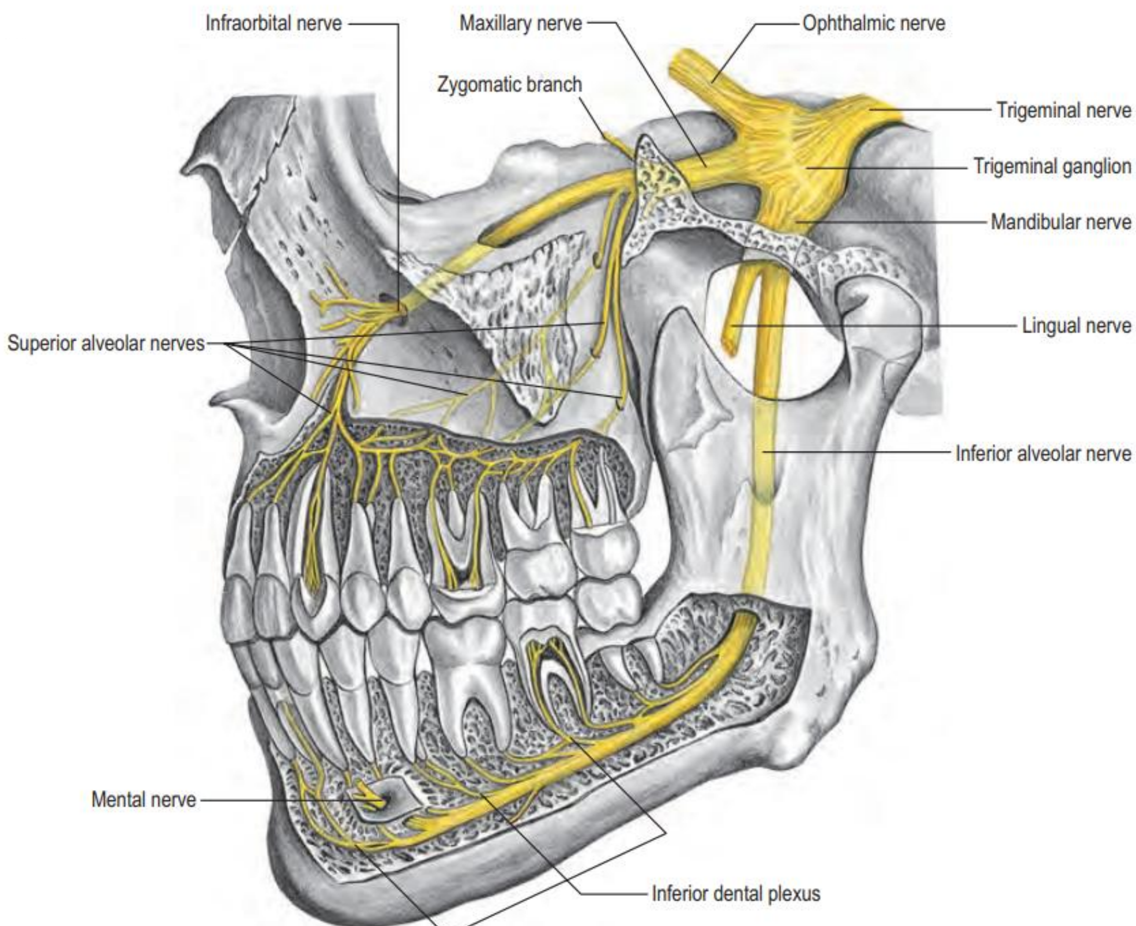
- Infraorbital
- Zygomatic
- Nasopalatine
- superior alveolar
- pharyngeal and
- greater palatine
- and lesser palatine



Maxillary nerve – branches

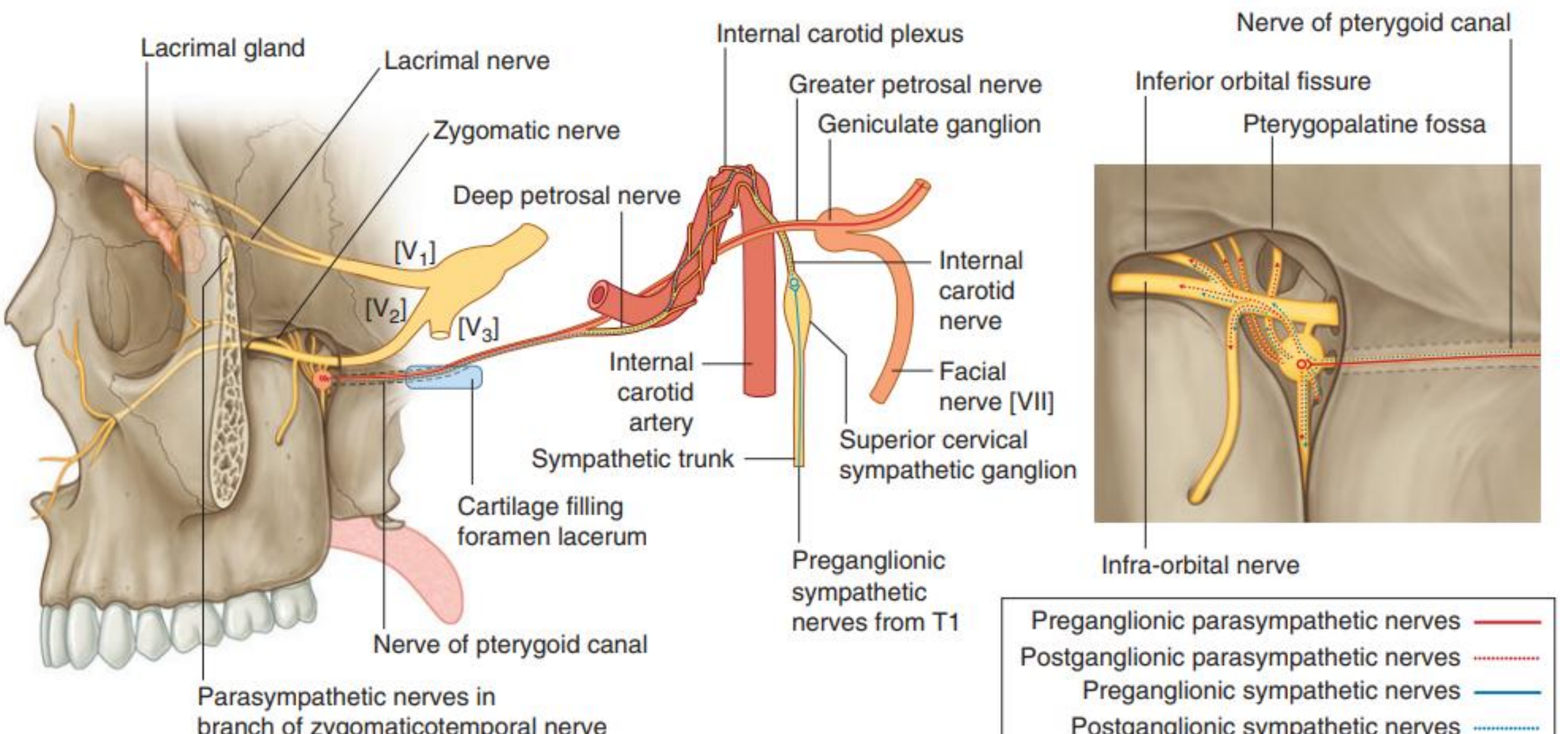
Superior alveolar (anterior, middle, posterior)

- Supplies the maxillary sinus and the upper molar teeth and adjoining parts of the gum and the cheek



Greater & Deep Petrosal Nerves

- Nerve of the pterygoid canal is formed by
  - greater petrosal nerve (branch of VII), which carries parasympathetic preganglionic fibers
  - deep petrosal nerve (branch of carotid plexus), which carries sympathetic postganglionic fibers
- Pterygopalatine ganglion receives fibers from the nerve of the pterygoid canal

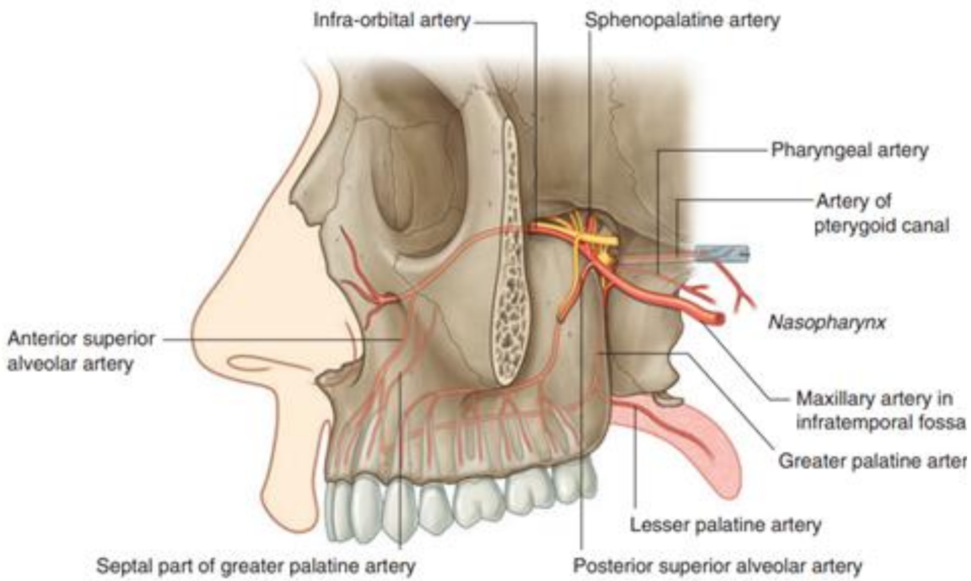


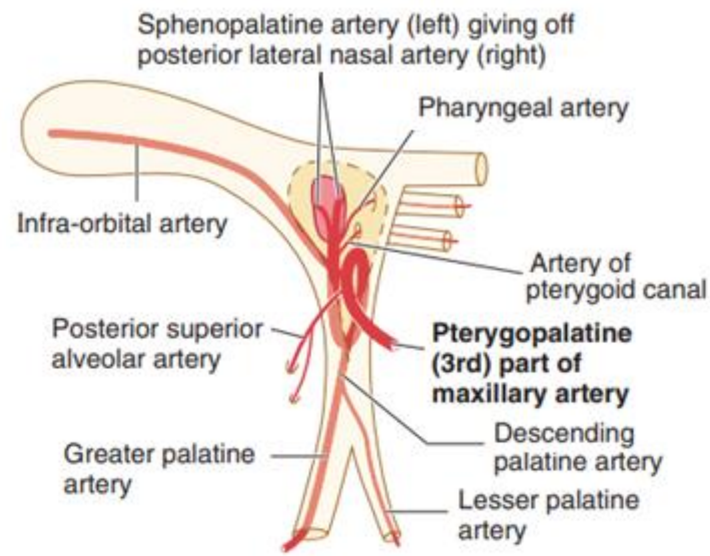
PT part of Maxillary artery

- Maxillary artery exits the IT fossa through the pterygomaxillary fissure to enter the PT fossa & follows the branches of the maxillary nerve (V_2)

These branches include (but not limited to):

- Infraorbital artery: lacrimal gland, some muscles of the eye
- Sphenopalatine artery: to the nasal cavity
- Descending palatine artery: branches into greater and lesser palatine arteries
- Posterior superior alveolar artery: to the teeth and gingiva

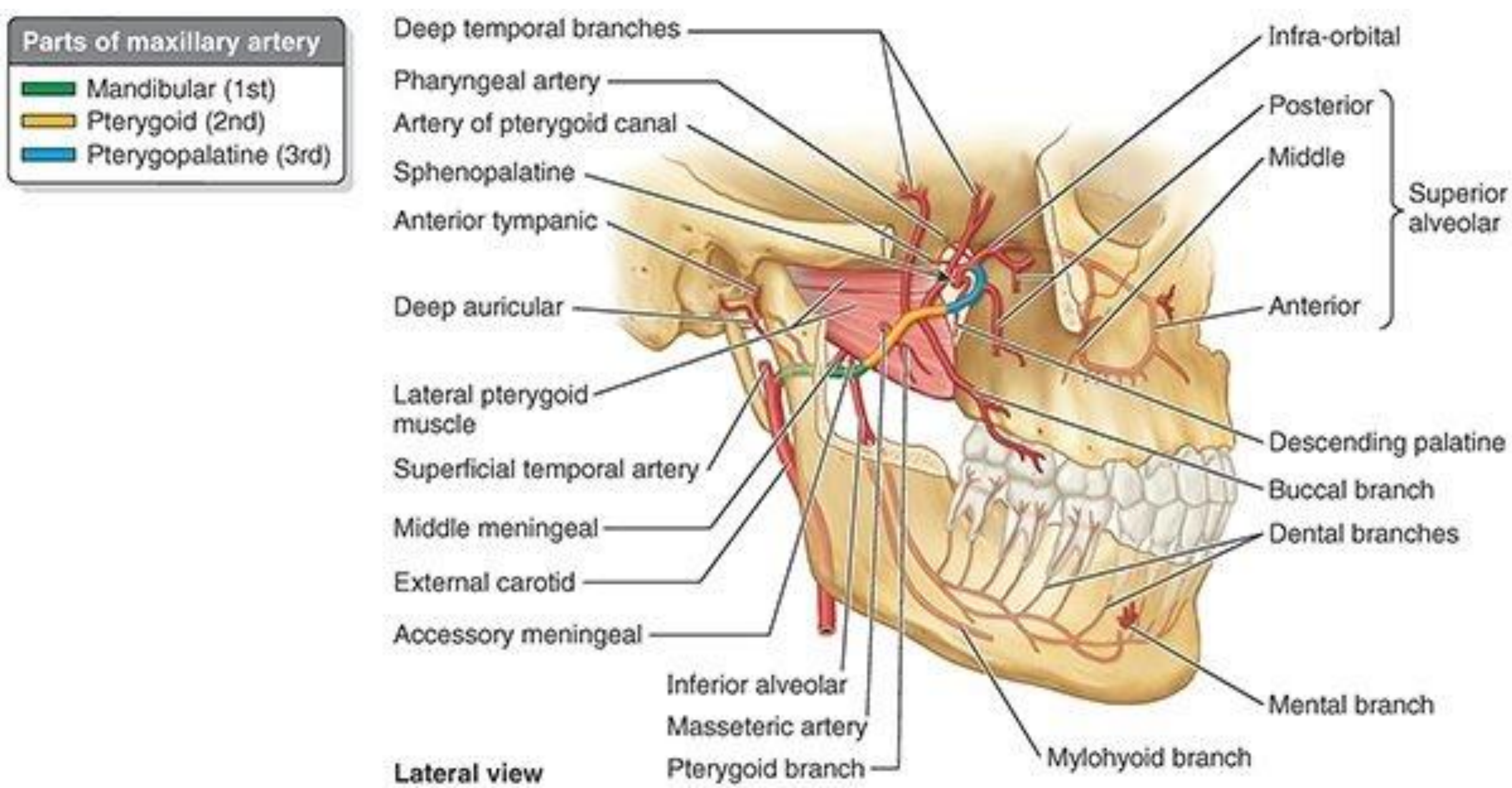




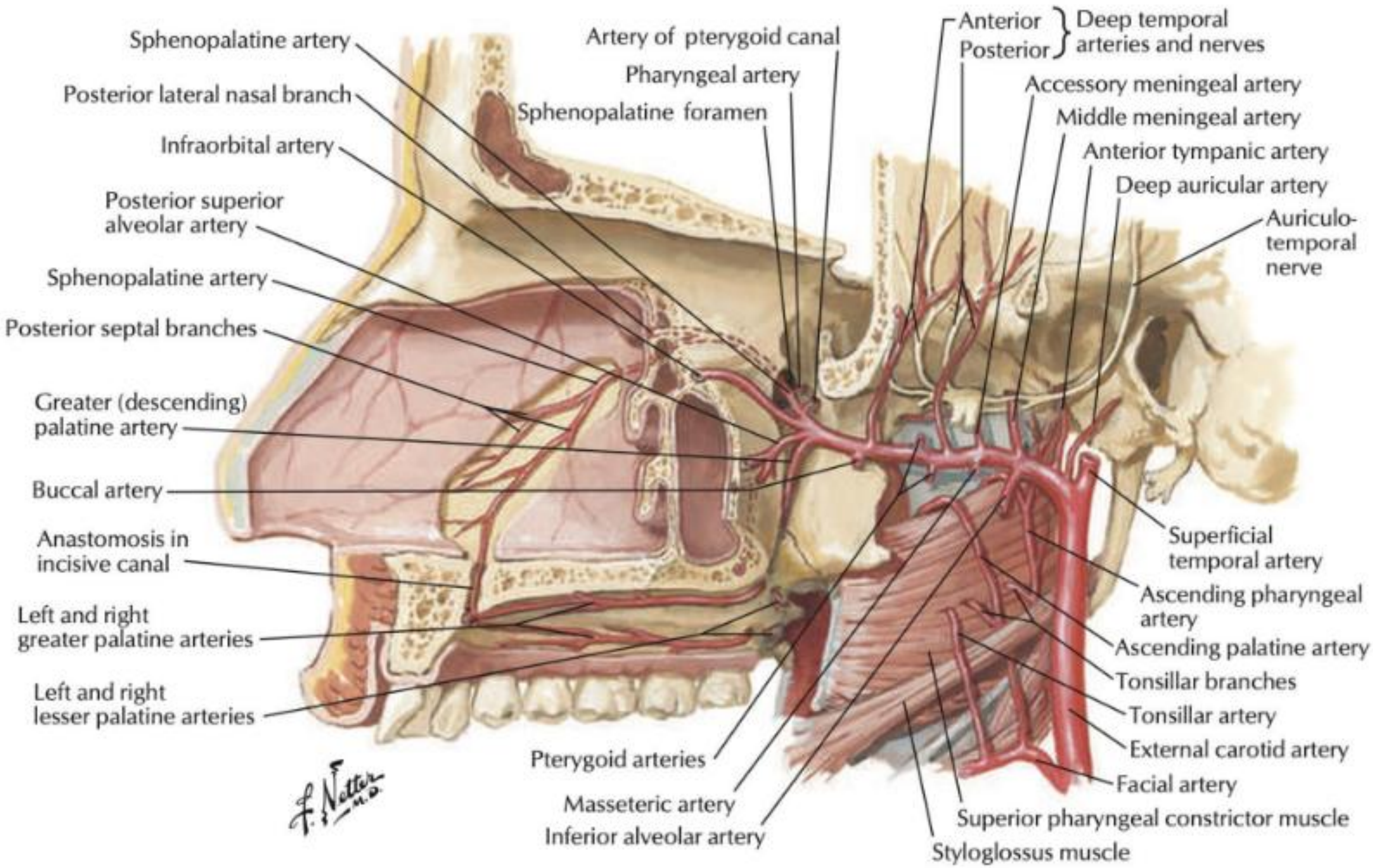
Maxillary artery divisions

3 portions by relation to the lateral pterygoid muscle

- 1st part (mandibular part): winds around deep neck of mandible
- 2nd part (pterygoid part): travels between 2 heads of lateral pterygoid
- 3rd part (pterygopalatine part): enters pterygopalatine fossa containing pterygopalatine ganglion



Summary – maxillary artery



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